

SAHIL IMTIYAZ

Postdoctoral Researcher, Basque Center for Applied Mathematics (BCAM) - Mathematical, Computational and Experimental Neuroscience Group
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WORK EXPERIENCE

- **Postdoctoral Researcher**, BCAM, Mathematical, Computational and Experimental Neuroscience Group, Spain (2024-2026).
- **Junior Scientist (affiliated)**, Canadian Quantum Research Center, British Columbia, Canada.

EDUCATION

- **PhD in Sciences and Technology: Computer Science**, University of Camerino, Italy. Defended 29 Nov 2023. Thesis: *Contextual Semantics Machinery*. Evaluation noted originality and very high quality in quantum contextuality, topology, and cohomological algebra.
- **MS in Computer Science**, International Islamic University Islamabad, Pakistan, 2018. Thesis: *Quantum Adiabatic Evolution for Global Optimization in Big Data*. Evaluation: 3.86/4.0.
- **Bachelor of Engineering in Information Technology**, Model Institute of Engineering and Technology, University of Jammu, India, 2014. Result: First Division.

SELECTED TALKS

- *Computation and Quantum Contextuality*, Bilkent University Quantum Computing Seminar, Ankara, 17 Oct 2025. [link](#).
- *The Gauge Theoretic Structure of Non-locality and Contextuality*, Istanbul Technical University, SIMMAG, 21 Oct 2025.
- *The Bundle-Theoretic Structure and Computational Modeling of Quantum Contextuality*, QNIP 2025, Indiana University, 6-9 Aug 2025. [link](#).
- *The Bundle-Theoretic Structure and Computational Modeling of Quantum Contextuality*, BYMAT, University of Seville, 17-20 Nov 2025. [abstracts](#).
- *Evolving Networks and Communication Systems*, Leioa Campus / Biofisika context, Spain, 28 Apr 2026.

TEACHING

- Theory of Computation and Distributed Coordination Calculus: course tutoring / teaching-assistant support during doctoral studies, University of Camerino; academic context: Prof. Emanuela Merelli.
- Artificial Intelligence and Data Mining Principles: master-level teaching support; academic context: Prof. Jamal Abdul Nasir.
- Physics and Mathematics: higher-secondary teaching experience with emphasis on foundations and laboratory-based learning.

ACADEMIC PROFILE

I am a mathematical and computational modeller working at the interface of quantum foundations, complex systems, and theoretical biology. My research develops structural frameworks for understanding how local constraints, contextual interactions, and evolving architectures give rise to global organisation, obstruction, and new forms of computation.

RESEARCH AREAS

Quantum foundations and contextuality; quantum information and computation; applied topology, geometry, and algebraic methods; topology-driven models of computation; evolving communication networks; mathematical and theoretical biology; biological communication and multi-level organisation; complex systems modelling.

SELECTED PUBLICATIONS

Peer-reviewed publications

1. S. Imtiyaz, E. Merelli and S. Rodrigues. *The Bundle-Theoretic Structure and Computational Modeling of Quantum Contextuality*. Springer CCIS 2659, 2026, pp. 274-301. [DOI](#).
2. S. Imtiyaz and S. Rodrigues. *Understanding Entanglement Through Quantifiable Algebraic Structures: Application to Bird Navigation*. IEEE TMBMC 12, 2026, pp. 279-297. [DOI](#).
3. M. Al Bataineh, T. Al Mallah, W. Mardini, S. Imtiyaz, Y. M. Khamayseh and M. B. Yasin. *Four-cycles Delay-Efficient RPL-based Routing Approach for IoT Applications*. ICICS 2021. [DOI](#).

Preprints

- S. Imtiyaz and S. Rodrigues. *The Gauge-Theoretic Structure of Non-Localities and Contextuality*. BCAM BIRD manuscript, 2025. [PDF](#).
- S. Imtiyaz. *Constraint Semantics for Multi-level Organisation*. bioRxiv preprint, 2026. [bioRxiv](#).
- S. Imtiyaz and E. Merelli. *Contextual Semantics Machinery*. BCAM manuscript, 2024. [PDF](#).
- S. Imtiyaz. *The Evolving Reconfigurable Network Bundle: A Geometric Framework for Evolving Communication Systems*. Under review, 2026.

Theses

- PhD thesis: *Contextual Semantics Machinery*, University of Camerino, 2023. [repository](#).
- MS thesis: *Quantum Adiabatic Evolution for Global Optimization in Big Data*, 2018. [arXiv](#).

AWARDS

- Italian PhD Scholarship (2018).
- Outstanding Achievement Award, CS-MAT Doctoral Curriculum, University of Camerino (2022).
- Best Poster Award, Computer Science and Mathematics Workshop, Italy (2022).
- National Eligibility Test, India (2010).

RESEARCH COLLABORATIONS DURING PHD

- Dalle Molle Institute for Artificial Intelligence, Lugano, Switzerland, with Andrea Danani.
- University of Oxford, Quantum Group, with Samson Abramsky.
- Ecole Normale Supérieure / CNRS Paris, with Giuseppe Longo.
- Basque Center for Applied Mathematics, with Serafim Rodrigues.

SUMMER SCHOOLS AND TRAINING

- Hilbert-Bernays Summer School on Logic and Computation, University of Göttingen, Germany. Project focus: Gödel incompleteness via the halting problem.
- 10th International ScaDS.AI Summer School, Leipzig, Germany, 2024.

SELECTED PROJECT / FUNDING CONTEXT

EU-FET TOPDRIM (FP7-ICT-318121); PID2023-146683OB-I00; BERC 2022-2025; BCAM Severo Ochoa CEX2021-001142-S; BCAM-IKUR; Bio-Molecules Computing Life BELL Project; GPS alarm scheduling; modular reconfigurable memories.

CORE COMPETENCIES

Theoretical framework development; literature synthesis and research positioning; structural formalisation; Python-based mathematical prototyping; LaTeX scientific writing; interdisciplinary teaching and presentation.